Laboratory work No. 5

Objective

The purpose of this paper is to introduce the concept of *function as a service* (FaaS)

Intro

"Azure Functions is a serverless cloud architecture that allows you to deploy and execute a piece of code without the need for a server infrastructure, web server or other configurations. In addition, Azure Functions can be used to write code in a variety of languages such as C#, Java, JavaScript, TypeScript and Python.

"Azure Functions is an event-driven, serverless computing platform used to run code snippets in Azure. "Azure Functions provides flexibility in deploying applications and building automation. If you are modernising your application from on-premises server infrastructure, consider moving to FaaS.

Serverless computing hosts and executes code in the cloud. It is "serverless" in the sense that no servers need to be installed or maintained to run the code.

The code runs on the servers of the cloud service provider. Developers write and deploy the code without worrying about the infrastructure needed to run it.

Typically, when a developer needs to run code, he needs to set up and maintain his own server. Configuring servers for applications is a time-consuming activity that is often not a developer's strong point.

Serverless computing or serverless architecture removes this infrastructure barrier for developers. The servers still run the code, but developers no longer have to worry about them. Developers can quickly deploy code without waiting for system administrators to install servers, patch operating systems or configure the network.

With serverless resources, you pay only for the computing resources needed to run the code. This consumption model can save costs because computing resources are used only when they are needed.

Now that you only pay for resources when you need them, you can quickly scale up or scale down volumes using autoscale. Autoscaling adds or removes computing resources to match changes in demand. Computing resources include elements such as processor, memory and network.

Azure Functions vs. Functions Apps vs. Functions

When working with Azure Functions, you'll see a lot of "features". This may seem confusing to a beginner. You need to know the names of the three "functions".

Azure Functions (capitalised) is the name of the service itself, for example Azure Automation, Azure Virtual Machines or Azure Storage Accounts.

Function App is a single application that uses Azure Functions.

Function - This is a fully code-defined function. You can have multiple functions (lower case) within a single Function App.

Triggers

"Azure Functions is "event-driven". This attribute specifies how the code executed by Azure Functions starts. A function starts with [some] event. That event can be many different things, each with a particular trigger.

A trigger is responsible for the execution of an Azure Function, and there are dozens of triggers to choose from. Here are some examples of events:

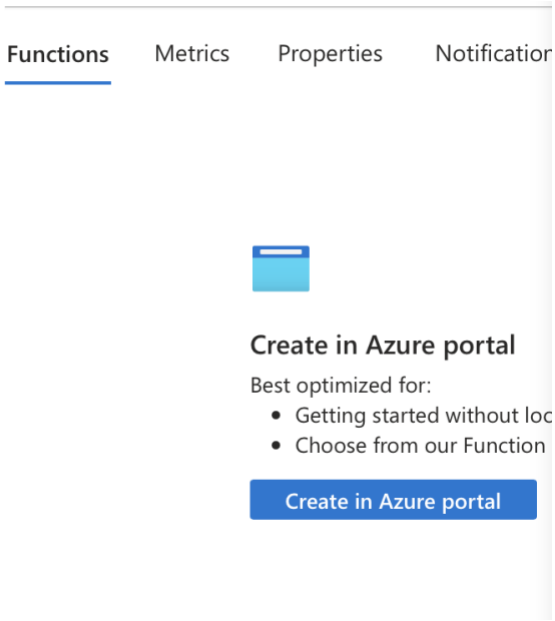
- HTTP trigger: Receiving an HTTP request, for example when interacting with an API to retrieve application information.
- Timer trigger: Time trigger: running on a schedule, for example sending reminders about upcoming meetings once a day.

Tasks

1. Log in to the Azure console
2. Select the Function App resource
3. Select a region, give it a name, etc.
4. You will need to select or create a storage account
5. Plan - Consumption
6. Enable monitoring - Insight

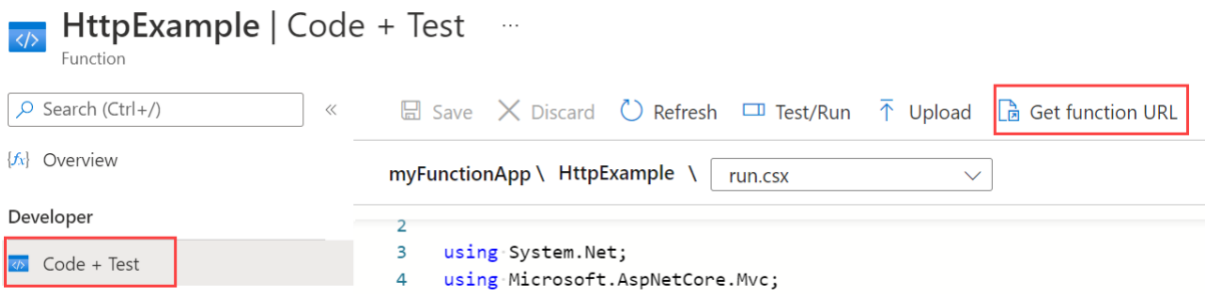
HTTP function

1. After creating a resource - go to the resource
2. In the Overview window, locate the Create in Azure portal



3. Click Create and select "Develop in portal"
4. Select "HTTP trigger"

5. Select the function name
6. Select the authorization level - Anonymous
7. In the next window. Find "Code + Test"
8. In the window, find the **Get function URL**



9. In the **Get function URL** window, select default and then copy the link.

10. Copy the link to your browser. Add an additional parameter ?name=<your name > at the end of the URL. What do you see? Can you tweak the code and come up with another attribute?
11. Find the **Log stream** menu, watch what happens when you call your function.
12. Show how the logs loo

Time function

1. Use the previous procedure, but this time choose Time Trigger instead of Http
2. In the "Code + Test" section, find the configuration and readme files.
3. Change the code to execute every 1 (one) minute
4. Go to Monitor -> Logs
5. Show that the function runs every minute.

In your report, please provide:

- Work progress
- Answer the questions asked in the course of the work